1. **TITLE: SIMPLE REFLEX AGENT FOR PARTICLE PICKING PROBLEM**

```
In [5]: %runfile D:/Suresh/code.py --wdir

    name: suresh dahal
    roll_no: 23
    lab_no:1

Start: A=Dirty, B=Dirty, Agent=A
Step 1: Percept=(A,Dirty) → Action=Clean → State[A=Clean, B=Dirty, Agent=A]
Step 2: Percept=(B,Clean) → Action=Move to B → State[A=Clean, B=Dirty, Agent=B]
Step 3: Percept=(B,Dirty) → Action=Clean → State[A=Clean, B=Clean, Agent=B]

SUCCESS: All clean!
```

2. **TITLE: UTILITY-BASED AGENT FOR SIMPLE WUMPUS WORLD**

```
In [19]: %runfile D:/Suresh/code.py --wdir

    name: suresh dahal
    roll_no: 23
    lab_no:2

Agent starts at A1

Step 1: At A1 | Breeze=True, Gold=False
Percept of breeze, A2 is risky, moving to B1 instead
Action: Move to B1

Step 2: At B1 | Breeze=False, Gold=False
No breeze → Safe to explore B2
Action: Move to B2

Step 3: At B2 | Breeze=True, Gold=True
Action: Grab gold

SUCCESS: Gold collected!

Actions taken: Move to B1, Move to B2, Grab gold
Result: SUCCESS
```

3. **TITLE: GOAL-BASED AGENT FOR SIMPLE WUMPUS WORLD**

```
In [25]: %runfile D:/Suresh/code.py --wdir

    name: suresh dahal
    roll_no: 23
    lab_no:3

Agent starts at A1

Step 1: Current Position = A1
  Possible moves: ['A2', 'B1']
  Evaluating A2 -> Wumpus
  Evaluating B1 -> Gold
Goal found at B1, moving there!

SUCCESS: Gold collected at B1!
```

4. **TITLE: MODEL-BASED AGENT IN A SIMPLE WUMPUS WORLD**

```
In [26]: %runfile D:/Suresh/code.py --wdir

    name: suresh dahal
    roll_no: 23
    lab_no:4

Plan to reach gold:
 - UP
 - RIGHT
 - RIGHT
 - UP
 - RIGHT
 - DOWN
 - GRAB
```

5. **TITLE: SOLVING 8-PUZZLE USING BFS ALGORITHM**

```
In [27]: %runfile D:/Suresh/code.py --wdir

    name: suresh dahal
    roll_no: 23
    lab_no:5

Steps to reach goal:
123
456
708

123
456
780
```

6. **TITLE: SOLVING 8-PUZZLE USING DFS ALGORITHM**

```
In [40]: %runfile D:/Suresh/code.py --wdir

    name: suresh dahal
    roll_no: 23
    lab_no:6

Steps to reach goal:
123
456
780
```

7. **TITLE: SOLVING 8-PUZZLE USING ITERATIVE DEEPENING DFS**

```
In [41]: %runfile D:/Suresh/code.py -

    name: suresh dahal
    roll_no: 23
    lab_no:7

Steps to reach goal:
123
406
758

123
456
708

123
456
780
```

8. **TITLE: PATH PROBLEM SOLUTION USING BEST FIRST SEARCH**

```
In [42]: %runfile D:/Suresh/code.py --wdir

    name: suresh dahal
    roll_no: 23
    lab_no:8

Path found: S → B → G
```

9. **TITLE: PATH PROBLEM SOLUTION USING A* SEARCH**

```
In [44]: %runfile D:/Suresh/code.py --wdir

    name: suresh dahal
    roll_no: 23
    lab_no:9

Shortest Path: ['A', 'C', 'D', 'E']
Total Cost: 9
```

10. **TITLE: BLOCK WORLD USING HILL CLIMBING SEARCH**

```
In [45]: %runfile D:/Suresh/code.py --wdir

    name: suresh dahal
    roll_no: 23
    lab_no:10

Steps to reach goal (Hill Climbing):
['C', 'A', 'B']
['A', 'C', 'B']
['A', 'B', 'C']

Goal reached!
```

11. TITLE: AND GATE USING HEBBIAN NEURAL NETWORK

```
In [46]: %runfile D:/Suresh/code.py --wdir

    name: suresh dahal
    roll_no: 23
    lab_no:11

After input [0, 0] → w1=0, w2=0, b=0
After input [0, 1] → w1=0, w2=0, b=0
After input [1, 0] → w1=0, w2=0, b=0
After input [1, 1] → w1=1, w2=1, b=1

Testing AND gate:
Input: (0, 0) → Output: 0
Input: (0, 1) → Output: 1
Input: (1, 0) → Output: 1
Input: (1, 1) → Output: 1
```

12. TITLE: AND GATE USING PERCEPTRON NEURAL NETWORK

```
    name: suresh dahal
    roll_no: 23
    lab_no:12

Epoch 1
Input: [0, 0], Target: 0, Output: 0, New w1=0, w2=0, b=0
Input: [0, 1], Target: 0, Output: 0, New w1=0, w2=0, b=0
Input: [1, 0], Target: 0, Output: 0, New w1=0, w2=0, b=0
Input: [1, 1], Target: 1, Output: 0, New w1=1, w2=1, b=1

Epoch 2
Input: [0, 0], Target: 0, Output: 1, New w1=1, w2=1, b=0
Input: [0, 1], Target: 0, Output: 1, New w1=1, w2=0, b=-1
Input: [1, 0], Target: 0, Output: 0, New w1=1, w2=0, b=-1
Input: [1, 1], Target: 1, Output: 0, New w1=2, w2=1, b=0

Epoch 3
Input: [0, 0], Target: 0, Output: 0, New w1=2, w2=1, b=0
Input: [0, 1], Target: 0, Output: 1, New w1=2, w2=0, b=-1
Input: [1, 0], Target: 0, Output: 1, New w1=1, w2=0, b=-2
Input: [1, 1], Target: 1, Output: 0, New w1=2, w2=1, b=-1

Epoch 4
Input: [0, 0], Target: 0, Output: 0, New w1=2, w2=1, b=-1
Input: [0, 1], Target: 0, Output: 0, New w1=2, w2=1, b=-1
Input: [1, 0], Target: 0, Output: 1, New w1=1, w2=1, b=-2
Input: [1, 1], Target: 1, Output: 0, New w1=2, w2=2, b=-1

Epoch 5
Input: [0, 0], Target: 0, Output: 0, New w1=2, w2=2, b=-1
Input: [0, 1], Target: 0, Output: 1, New w1=2, w2=1, b=-2
Input: [1, 0], Target: 0, Output: 0, New w1=2, w2=1, b=-2
Input: [1, 1], Target: 1, Output: 1, New w1=2, w2=1, b=-2

Testing the trained Perceptron:
Input: (0, 0) → Output: 0
Input: (0, 1) → Output: 0
Input: (1, 0) → Output: 0
Input: (1, 1) → Output: 1
```

13. TITLE: BACKPROPAGATION NEURAL NETWORK TRAINING ALGORITHM

```
In [48]: %runfile D:/Suresh/code.py --wdir

    name: suresh dahal
    roll_no: 23
    lab_no:13

Epoch 0: Error = 0.6942
Epoch 1000: Error = 0.0770
Epoch 2000: Error = 0.0340
Epoch 3000: Error = 0.0211
Epoch 4000: Error = 0.0151
Input: 0, 0 → Output: 0.081
Input: 0, 1 → Output: 0.950
Input: 1, 0 → Output: 0.949
Input: 1, 1 → Output: 1.000
```

14. TITLE: GENETIC ALGORITHM IMPLEMENTATION

```
In [49]: %runfile D:/Suresh/code.py --wdir

    name: suresh dahal
    roll_no: 23
    lab_no:14

Initial Population: ['00011', '10100', '10100', '01001', '00101', '10101']
Gen 1: Best = 11100 (28), Fitness = 784
Gen 2: Best = 11101 (29), Fitness = 841
Gen 3: Best = 11111 (31), Fitness = 961
Gen 4: Best = 11101 (29), Fitness = 841
Gen 5: Best = 11101 (29), Fitness = 841
Gen 6: Best = 11101 (29), Fitness = 841
Gen 7: Best = 11110 (30), Fitness = 900
Gen 8: Best = 11110 (30), Fitness = 900
Gen 9: Best = 11110 (30), Fitness = 900
Gen 10: Best = 11111 (31), Fitness = 961

Best Solution: 11111 (x = 31) → f(x) = 961
```

15. TITLE: SPAM DETECTION USING NAIVE BAYES CLASSIFIER

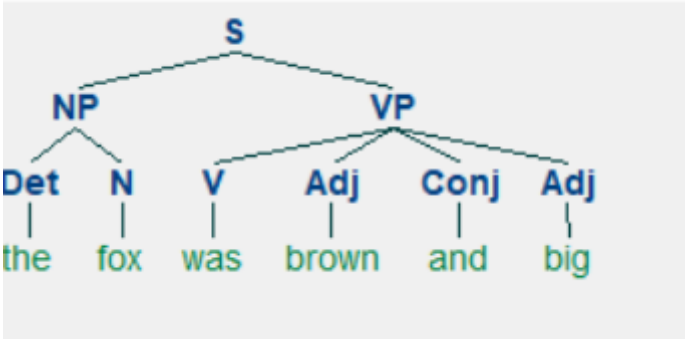
```
In [50]: %runfile D:/Suresh/code.py --wdir

    name: suresh dahal
    roll_no: 23
    lab_no:15

SPAM DETECTION RESULTS:

Message: 'buy offer now' → SPAM
Message: 'hello offer' → HAM
Message: 'meet friend tomorrow' → HAM
Message: 'cheap prize' → SPAM
```

16. TITLE: PARSE TREE GENERATION OF A SENTENCE




```
In [5]: %runfile D:/Suresh/code.py --wdir

    name: suresh dahal
    roll_no: 23
    lab_no:16

Enter a sentence: the fox was brown and big

Parse Trees:

(S
  (NP (Det the) (N fox))
  (VP (V was) (Adj brown) (Conj and) (Adj big)))
```